

Demand System

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- This presentation refers to chapter 2 (2.1, 2.2, 2.3.1, 2.3.2, 2.3.7) of **Empirical Industrial Organization: Models, Methods, and Applications** by Victor Aguirregabiria.

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space

Table of Contents

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space

Why do we care about demand curve?

- Products in almost all markets are differentiated to some extent.
- Products differ in their physical characteristics, location, etc. This is true even for markets that are seemingly homogeneous (e.g. drugs, gasoline).
- We need some knowledge of the demand system in order to answer a broad set of questions. For example:
 - Measuring market power
 - Assessing the welfare effects of mergers
 - Assessing the effects of new taxes or tariffs
 - Analyzing product introduction
 - Estimating the welfare effects of new goods
- We may also be interested in consumer behavior per se: who buys what products, and the importance of search costs, switching costs, consumer information, advertising brand loyalty and so forth for purchasing behavior.

Some preliminary comments

- Theory on differentiated products (Anderson, de Palma, and Thisse; Shaked and Sutton; Caplin and Nalebuff; and others) is mainly focused on the supply side: given a demand system, what are the products and prices chosen in equilibrium?
- Due to various reasons (data, generality, and complexity) the empirical literature has tended to focus more on the demand side.
- Note that in many regulated industries (electricity, water, postal services) prices are set through a non-market mechanism, making the demand side the natural place to start.

Some preliminary comments

- The individual choice aspect of demand often leads to introspection and a tendency to make the models richer and “more realistic”. We should never forget that there is a clear trade-off!
- Think about adding richness from a cost-benefit analysis: richer models are typically more complicated to compute, test, and to communicate to others. These costs should be compared to the added-value the complexity buys us. If this value is small, we may be better off with a simpler model.
- You will also see that many demand papers focus a lot on specifying a theoretically appealing econometric model, and often put less emphasis on whether the data have the relevant variation to identify the key parameters.
- Remember that empirical models should always be evaluated in light of the question at hand and the data available to answer it.

Table of Contents

- 1 Motivation
- 2 Single product model**
- 3 Basic Assumption
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space

Single Product Model: the Simplest Approach

- Suppose we observe product i across markets or time periods t : quantities q_{it} and prices p_{it} .
- The simplest approach: assume a **single demand curve**, common across products, that depends only on **own price**.
- Two standard specifications:

1. Level–Level regression:

$$q_{it} = \beta_0 + \beta_1 p_{it} + \varepsilon_{it}$$

2. Log–Log regression:

$$\log(q_{it}) = \beta_0 + \beta_1 \log(p_{it}) + \varepsilon_{it}$$

Interpreting β_1 : Price Elasticity of Demand

- In the **log–log** specification, β_1 is the **price elasticity of demand**:

$$\beta_1 = \frac{\partial \log(q_{it})}{\partial \log(p_{it})} = \frac{\partial q_{it}/q_{it}}{\partial p_{it}/p_{it}}$$

- We expect $\beta_1 < 0$: a 1% increase in price leads to a $|\beta_1|\%$ decrease in quantity demanded.
- In the **level–level** specification, $\beta_1 = \partial q_{it}/\partial p_{it}$, which depends on the units of measurement – less convenient for comparing across products or markets.
- The log–log specification also has a natural micro foundation: it is the demand equation implied by a **Cobb–Douglas** utility function (as we derived in class).

Single Product Model: Endogeneity

- Prices and quantities are **simultaneously determined** by the equilibrium of supply and demand. This implies:

$$E[\varepsilon_{it} | \log(p_{it})] \neq E[\varepsilon_{it}] = 0$$

- So, the OLS estimator is *biased* and *inconsistent*.
- The bias can be eliminated or, at least, attenuated by:
 - Control function:** adding exogenous controls:

$$\log(q_{it}) = \beta_0 + \beta_1 \log(p_{it}) + \mathbf{X}_{it}'\gamma + \varepsilon_{it}$$

- Individual and time fixed effects:**

$$\log(q_{it}) = \beta_0 + \beta_1 \log(p_{it}) + \eta_i + \lambda_t + \varepsilon_{it}$$

- Instrumental variables:** finding an instrument Z_{it} for prices, such that:

$$\log(p_{it}) = \alpha_0 + \alpha_1 Z_{it} + \zeta_{it} \quad \log(q_{it}) = \beta_0 + \beta_1 \widehat{\log(p_{it})} + \varepsilon_{it}$$

Single Product Model: What is a Good Instrument?

- Recall the two conditions for a valid instrument Z_{it} :
 - ① **Relevance**: $\text{Cov}(Z_{it}, \log(p_{it})) \neq 0$
 - ② **Exogeneity**: $\text{Cov}(Z_{it}, \varepsilon_{it}) = 0$
- Economically: a good instrument is something that **shifts the supply curve** without shifting the demand curve.
- The natural candidates are **cost shifters**:
 - Input prices (wages, raw material costs, energy prices)
 - Transportation costs
 - Taxes on inputs

Limitations of the Single Product Model

- The single product model is simple and tractable, but has a **fundamental limitation**: it ignores **cross-price effects**.
 - In general, the demand for product j depends on the prices of other products $k \neq j$ – whether they are **substitutes** or **complements**. The single product model cannot capture either.
 - Example: if Coca-Cola raises its price, demand for Pepsi increases (substitutes); but demand for ice may also change (complements). Both effects are invisible to the single product model.
 - We need a richer framework: a **demand system** that models the demand for *all* products jointly.
- ⇒ Before building demand systems, we need to set up the **consumer's optimization problem** more carefully.

Table of Contents

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption**
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space

Why Can't We Estimate Preferences Directly?

- In the production function, output Y is observable. We could write $\log Y = \alpha \log K + \beta \log L + \varepsilon$ and run a regression directly.
- Here, the “output” of the consumer’s problem is **utility** U – which we never observe.
- We need an **indirect approach**: instead of estimating utility directly, we estimate **demand equations** – i.e., we measure how changes in prices affect the quantities purchased.
- This approach rests on a key assumption: the consumer bought the observed bundle because it **maximized her utility** given prices and income. The quantities we observe in the data therefore *reveal* her preferences – this is the principle of **revealed preferences**.

Consumer Preferences: Setup

- There are J products, indexed by $j \in \{1, 2, \dots, J\}$. Let q_j be the quantity of product j purchased.
- We introduce an **outside good** c : the composite of all goods not in $\{1, \dots, J\}$. Its price is normalized to 1, so c represents the dollar expenditure on outside goods.
- The consumer has a utility function defined over all goods:

$$U(c, q_1, q_2, \dots, q_J)$$

Note: We cannot observe U directly – unlike Y in the production function setting.

Consumer Problem

- The consumer maximizes utility subject to the budget constraint:

$$\max_{\{c, q_1, \dots, q_J\}} U(c, q_1, q_2, \dots, q_J)$$

$$\text{subject to: } c + p_1 q_1 + p_2 q_2 + \dots + p_J q_J \leq y$$

- where p_j is the price of product j and y is disposable income.
- The price of c is normalized to 1, so the constraint is linear in all choice variables.

Lagrangian and First Order Conditions

$$\max_{\{c, q_1, \dots, q_J\}} U(c, q_1, \dots, q_J) + \lambda[y - c - p_1 q_1 - \dots - p_J q_J]$$

- The first order conditions are:

$$\begin{cases} U_j - \lambda p_j = 0 & \text{for } j = 1, 2, \dots, J \\ U_0 - \lambda = 0 \\ y - c - p_1 q_1 - \dots - p_J q_J = 0 \end{cases}$$

- where $U_j \equiv \partial U / \partial q_j$ is the marginal utility of product j , and $U_0 \equiv \partial U / \partial c$.
- The FOC for the outside good gives $\lambda = U_0$: the Lagrange multiplier equals the marginal utility of income.

Marshallian Demand Equations

- Solving the system of FOCs gives the **Marshallian demand equations** – one for each product:

$$q_1 = f_1(p_1, p_2, \dots, p_J, y)$$

$$q_2 = f_2(p_1, p_2, \dots, p_J, y)$$

$$\vdots$$

$$q_J = f_J(p_1, p_2, \dots, p_J, y)$$

- Each f_j depends on **all prices** and income – capturing both own-price and cross-price effects.
- This is exactly what was missing from the single product model.

Properties of a Demand System

- Not every system relating quantities and prices is a valid demand system. It must arise as the solution to a consumer optimization problem. Two necessary properties:
- **Adding up:**

$$c + \sum_{j=1}^J p_j f_j(p, y) = y$$

Total expenditure exhausts income.

- **Homogeneity of degree zero:**

$$f_j(\delta p, \delta y) = f_j(p, y) \quad \forall \delta > 0$$

Only relative prices matter – no money illusion.

Note: These properties are automatically satisfied when demand is derived from utility maximization – but not by arbitrary regression equations.

What Makes a Good Demand System? - (1)

Why not just run J separate regressions of q_j on all J prices?

- **Dimensionality.** With J products, each equation has J price regressors – that is J^2 parameters to estimate with no structure linking them. With large J this quickly becomes infeasible.
- **No guarantee of rationalizability.** Nothing ensures that J separate regressions satisfy adding up or homogeneity of degree zero – the system may not correspond to any utility function. Without that, welfare analysis and structural inference are invalid.

What Makes a Good Demand System? - (2)

Deriving demand from a utility function solves both problems. The empirical literature looks for utility functions generating demand systems with two properties:

- 1 **Simple enough** to be estimable with standard econometric methods (e.g. linear regression).
- 2 **Flexible enough** to allow for rich patterns of substitution and complementarity across products.

There is a fundamental **trade-off** between the two. We will study three demand systems in product space that represent different points on it.

Product vs. Characteristic space

There are two general approaches to estimating demand:

- **Product space** is more natural: consumers have preferences over products, and those preferences lead to demand at the product level.
- The **characteristic space** approach views products as bundles of characteristics, and consumers having preferences over those characteristics, rather than over the bundles.

Table of Contents

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption
- 4 Demand systems in product space**
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space

Demand systems in product space

For consumer demand models in product space, we are going to consider three different demand systems.

- 1 Linear expenditure (or Stone-Geary) demand system.
- 2 Constant Elasticity of Substitution (CES) demand system.
- 3 Deaton-Muellbauer (or 'Almost Ideal') demand system.

Linear Expenditure (Stone-Geary) Demand System

- Consider the utility function:

$$U = c(q_1 - \gamma_1)^{\alpha_1}(q_2 - \gamma_2)^{\alpha_2} \cdots (q_J - \gamma_J)^{\alpha_J}$$

where α 's and γ 's are positive parameters.

- $\gamma_j \geq 0$ can be interpreted as the minimum amount of consumption of good j that a consumer needs to “survive” ($U > 0$).
- $\alpha_j > 0$ represents the “intensity” of product j in generating utility.
- The marginal utilities are:

$$U_j = \alpha_j \frac{U}{q_j - \gamma_j} \quad \text{and} \quad U_0 = \frac{U}{c}$$

- Note that all the products are complements in this utility function:

$$U_{jk} = \alpha_j \alpha_k \frac{U}{(q_j - \gamma_j)(q_k - \gamma_k)} > 0$$

Linear Expenditure Demand System [2]

- $U_0 - \lambda = 0$ implies that $\frac{U}{c} - \lambda = 0$, and $c = \frac{U}{\lambda}$
- $U_j - \lambda p_j = 0$ implies that $\alpha_j \frac{U}{q_j - \gamma_j} = \lambda p_j$, and $\alpha_j \frac{1}{q_j - \gamma_j} \frac{U}{\lambda} = p_j$.
- Combining these equations, we obtain the **Linear Expenditure System**:

$$q_j = \gamma_j + \alpha_j \frac{c}{p_j} \text{ for } j = 1, 2, \dots, J$$

$$\text{and } c = \frac{y - \sum_{j=1}^J p_j \gamma_j}{1 + \sum_{j=1}^J \alpha_j}$$

- Note that for any $j \neq k$, we have that:

$$\frac{\partial q_j}{\partial p_k} = \frac{\alpha_j}{p_j} \frac{\partial c}{\partial p_k} = \frac{-\alpha_j \gamma_j}{p_j \left[1 + \sum_{i=1}^J \alpha_i \right]} < 0$$

All the products are complements.

Linear Expenditure Demand System [3]

- It is very convenient because its simplicity.
- Suppose that we have data of individual purchases and prices over T periods of time ($t = 1, 2, \dots, T$) : $\{c_t, q_{1t}, q_{2t}, \dots, q_{Jt}\}$ and $\{p_{1t}, p_{2t}, \dots, p_{Jt}\}$.
- The model implies a system of J linear regressions. For product j :

$$q_{jt} = \gamma_j + \alpha_j x_{jt} + \xi_{jt}$$

with $x_{jt} = c_t/p_{jt}$, and the error term ξ_{jt} could be measurement error in q_{jt} .

- But it is very restrictive. It imposes the restriction that **all the goods are complements in consumption**.
- This is not realistic in most applications, particularly when the goods under study are varieties of a differentiated product.

CES Demand System

- Consider the CES utility function:

$$U = c \left(\sum_{j=1}^J q_j^\sigma \right)^{1/\sigma}$$

where $\sigma \in [0, 1]$ is a parameter that represents the degree of substitution between the J products.

- The marginal utilities are:

$$U_j = q_j^{\sigma-1} \quad \text{and} \quad U_0 = \frac{U}{c}$$

where $Q_\Sigma \equiv \sum_{i=1}^J q_i \sigma$.

- It is simple to verify that for any two pairs of products, j and k , we have that $\frac{\partial^2 U}{\partial q_j \partial q_k} \equiv \frac{\partial U_j}{\partial q_k} < 0$. **All the products are substitutes in consumption.**

CES Demand System [2]

- $U_0 - \lambda = 0$ implies that $\frac{U}{c} - \lambda = 0$, and $c = \frac{U}{\lambda}$
- $U_j - \lambda p_j = 0$ implies that $q_j^{\sigma-1} \frac{U}{Q_\Sigma} = \lambda p_j$, and $\frac{q_j^{\sigma-1}}{Q_\Sigma} \frac{U}{\lambda} = p_j$.
- Combining these equations, we the CES System:

$$q_j = \left[\frac{c}{p_j Q_\Sigma} \right]^{\frac{1}{1-\sigma}} \quad \text{for} \quad j = 1, 2, \dots, J$$

- For any pair of products, j and k , we have that:

$$\log \left(\frac{q_j}{q_k} \right) = \frac{-1}{1-\sigma} \log \left(\frac{p_j}{p_k} \right)$$

CES Demand System [3]

- The CES is also very convenient because its simplicity.
- Suppose that we have data of individual purchases and prices over T periods of time: $\{c_t, q_{1t}, q_{2t}, \dots, q_{Jt}\}$ and $\{p_{1t}, p_{2t}, \dots, p_{Jt}\}$.
- The model implies a system of J linear regressions. For product $j > 1$:

$$\log(q_{jt}/q_{1t}) = \beta \log(p_{jt}/p_{1t}) + \xi_{jt}$$

where $\beta = \frac{-1}{1-\sigma}$, and the error term ξ_{jt} could be measurement error in $\log q_{it}$.

- But it is very restrictive. It imposes the restriction that **the substitution between any pair of products is exactly the same.**

Demand System [4]

- The CES system imposes strong restrictions on cross-price elasticities. For any three varieties, say j , k , and l :

$$Elasticity_{k,j} = \frac{d \log q_k}{d \log p_j} = \frac{d \log q_l}{d \log p_j} = Elasticity_{l,j}$$

- Suppose that we use this system to study the demand of different varieties of automobiles.
- Suppose that products j and k are “similar” luxury cars, and product l is an basic and expensive variety.
- The CES model implies that a reduction in the price of the luxury car j implies the same proportional increase in the demand of the other luxury car k and the basic car l .
- This is very unrealistic.

Deaton-Muellbauer ('Almost Ideal') Demand System

- They propose the utility function:

$$U = c \left[\prod_{j=1}^J (q_j - \gamma_j)^{\alpha_j} \right] + \delta_{jk} \sum_{j=1}^J \sum_{k=1}^J q_j q_k$$

where γ 's, α 's and δ 's are parameters. α 's and γ 's are positive, but δ 's can be positive or negative.

- This utility allows for complementarity and substitutability between products, and for a flexible pattern of substitution between different products.
- The marginal utilities are:

$$U_j = \alpha_j \frac{U^*}{q_j - \gamma_j} + \sum_{k=1}^J \delta_{jk} q_k \text{ and } U_0 = \frac{U^*}{c}$$

where $U^* \equiv c \prod_{i=1}^J (q_i - \gamma_i)^{\alpha_i}$

'Almost Ideal' Demand System [2]

- For any pair of products, j and k , we have that:

$$U_{jk} = \alpha_j \alpha_k \frac{U^*}{(q_j - \gamma_j)(q_k - \gamma_k)} + \delta_{jk} \begin{matrix} \geq \\ \leq \end{matrix} 0$$

that can be positive (complements) or negative (substitutes) and take different values for each pair of products.

'Almost Ideal' Demand System [3]

- $U_0 - \lambda = 0$ implies that $\frac{U^*}{c} - \lambda = 0$, and $c = \frac{U^*}{\lambda}$.
- $U_j - \lambda p_j = 0$ implies that $\alpha_j \frac{U^*}{q_j - \gamma_j} + \sum_{k=1}^J \delta_{jk} q_k - \lambda p_j = 0$, and $\alpha_j \frac{c}{q_j - \gamma_j} + \sum_{k=1}^J \delta_{jk}^* q_k = p_j$.
- Combining these equations, it is possible to derive the following system of Marshallian demand equations:

$$w_j = \beta_j^{(0)} + \beta_j^{(y)} \log(y) + \sum_{k=1}^J \beta_{jk}^{(p)} \log(p_k)$$

- $w_j \equiv \frac{p_j q_j}{y}$ is the expenditure share of product j ;
- $\{\beta_j^{(0)}, \beta_j^{(y)}, \beta_{jk}^{(p)}\}$ are parameters which are known functions of the utility parameters $\{\alpha_j, \gamma_j, \delta_{jk}\}$.

'Almost Ideal' Demand System [4]

$$w_j = \beta_j^{(0)} + \beta_j^{(y)} \log(y) + \sum_{k=1}^J \beta_{jk}^{(p)} \log(p_k)$$

- The model implies the symmetry conditions $\beta_{jk}^{(p)} = \beta_{kj}$ (Slutsky's symmetry condition).
- Therefore, the number of free parameters is: $2J + \frac{J(J+1)}{2}$, that increases quadratically with the number of products.

'Almost Ideal' Demand System [5]

- Suppose that we have data on individual purchases, income, and prices over T periods of time: $\{c_t, q_{1t}, q_{2t}, \dots, q_{Jt}\}, y_t$, and $\{p_{1t}, p_{2t}, \dots, p_{Jt}\}$.
- For each product j , we can estimate the regression equation:

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \log(y_t) + \beta_{j1}^{(p)} \log(p_{1t}) + \dots + \beta_{jJ}^{(p)} \log(p_{Jt}) + \xi_{jt}$$

where the error ξ_{jt} can be interpreted as measurement error in w_{jt} .

Multi-stage Budgeting

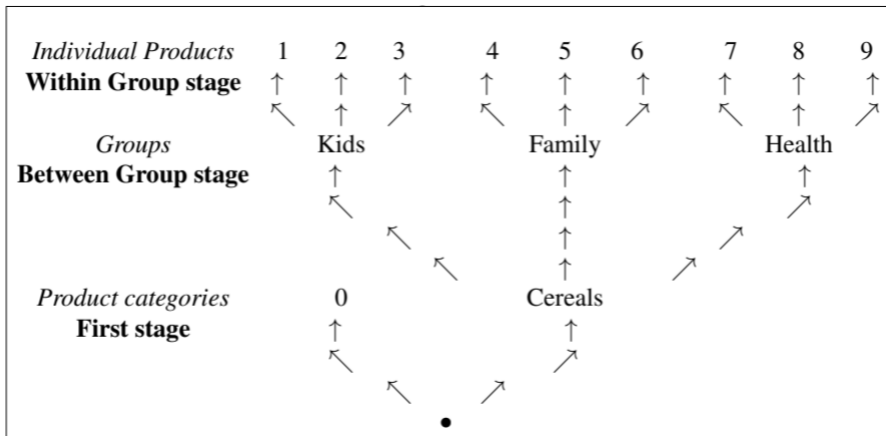
- For products with many (> 100) varieties (automobiles, smartphones, cereals, beer, etc) the number of parameters to estimate in the AI demand system can be very large, even larger than the #observations.
- Deaton and Muellbauer propose using a multi-stage budgeting approach.
- Suppose that the utility function is separable in the utility from G groups of products:

$$U = v_1(\tilde{\mathbf{q}}_1) + v_2(\tilde{\mathbf{q}}_2) + \cdots + v_G(\tilde{\mathbf{q}}_G)$$

$\tilde{\mathbf{q}}_g$ = Vector of quantities of varieties in group g ;

$v_g(\tilde{\mathbf{q}}_g)$ = Sub-utility from group g

Multi-stage Budgeting



Multi-stage Budgeting (under AIDS)

- Then, the demand system at the lower stage (within-group stage) is:

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \log \left(\frac{y_{gt}}{P_{gt}} \right) + \sum_{k \in \mathcal{J}_g} \beta_{jk}^{(p)} \log(p_{kt})$$

y_{gt} = Expenditure in group g ;

P_{gt} = Price index for group g .

- According to the model, this price index depends on the parameters of the model in group g . Non-linear system. Typically applications use “short-cuts”: e.g., $\log P_{gt} = \sum_{j \in \mathcal{J}_g} w_{jt} \log(p_{jt})$.
- Number of parameters increases quadratically with J_g but not with J .

Multi-stage Budgeting (under AIDS) [2]

- The demand system at the **group stage** is:

$$\frac{y_{gt}}{y_t^*} = \beta_g^{(0,2)} + \beta_g^{(y,2)} \log\left(\frac{y_t^*}{P_t^*}\right) + \sum_{g'=1}^G \beta_{g,g'}^{(p,2)} \log(P_{gt})$$

y_t = Total expenditure in the large category (e.g., cereals);

P_t = Price index for the large category (e.g., cereals).

- Finally, at the top-stage, the demand for the category is:

$$\frac{y_t^*}{y_t} = \beta^{(0,3)} + \beta^{(y,3)} \log(y_t) + \beta^{(p,3)} \log(P_t^*)$$

Data

- Ideally, we would like to have data on consumer decisions at the consumer or household level.
- This type of data is available from consumer surveys but at the level of large product categories:
 - e.g., clothing (different types), food (different types).
- In Empirical IO, we are interested in estimating demand system between very specific type of products, for example:
 - the cross demand elasticity between iPhone-15 and iPhone-16; or between iPhone-16 and Samsung Galaxy;
 - or between a Toyota Corolla and VW Jetta (or for that matter, with any other automobile product).
- For these specific purchases, consumer level data is not so commonly available.

Data [2]

- The most common type of data available for the estimation of demand systems is aggregate market level data.
- Data on aggregate quantities and prices from all the consumers in a particular period t (year, month):

$$q_{jt}, p_{jt} \text{ for } j = 1, 2, \dots, J \text{ and } t = 1, 2, \dots, T$$

- In some cases, we have this information for M different geographic markets (e.g., cities), that we index by m :

$$q_{jmt}, p_{jmt} \text{ for } j = 1, 2, \dots, J, t = 1, 2, \dots, T, \text{ and } m = 1, 2, \dots, M$$

- This is the type of data that we will consider for most of this topic.

Estimation: OLS

- Given the linear regression model:

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \ln(y_t) + \beta_{j1}^{(p)} \ln(p_{1t}) + \cdots + \beta_{jJ}^{(p)} \ln(p_{Jt}) + \xi_{jt}$$

- We can estimate the model parameters using OLS, separately for each product j (using the T observations for this product).
- Under the assumption that the error term ξ_{jt} is just measurement error in the dependent variable w_{jt} , and this measurement error is not correlated with prices or income, then the OLS estimator is consistent (unbiased as T is large).
- However, these conditions are very restrictive/unrealistic.

Endogeneity (Simultaneity)

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \ln(y_t) + \beta_{j1}^{(p)} \ln(p_{1t}) + \cdots + \beta_{jJ}^{(p)} \ln(p_{Jt}) + \xi_{jt}$$

- At least part of the error term ξ_{jt} represents consumers' taste/preference for product j that is known to consumers but unknown to us as researchers.
- Advertising, promotions, changes in taste over time, and changes in utility parameters are behind the error term ξ_{jt} .
- We will represent all these factors under the term: unobserved demand shock (for product j).
- This unobserved demand shock is known to consumers but unknown to us as researchers.

Endogeneity (Simultaneity)

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \ln(y_t) + \beta_{j1}^{(p)} \ln(p_{1t}) + \dots + \beta_{jJ}^{(p)} \ln(p_{Jt}) + \xi_{jt}$$

- The unobserved demand shock (or part of it) can be also known by firms selling product j .
- Therefore, they will take into account ξ_{jt} when making their pricing decision.
- A higher ξ_{jt} implies a higher Marginal Revenue and therefore a higher price.
- We expect ξ_{jt} to be correlated with $\ln(p_{jt})$.
- The OLS estimator is inconsistent because the regressors are correlated with the error term.

Endogeneity (Simultaneity)

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \ln(y_t) + \beta_{j1}^{(p)} \ln(p_{1t}) + \dots + \beta_{jJ}^{(p)} \ln(p_{Jt}) + \xi_{jt}$$

- We expect prices of products other than j also being correlated with ξ_{jt} .
- This is (mainly) because the demand shocks ξ_s of the different products can be correlated with each other: ξ_{kt} is correlated with p_{kt} , and ξ_{kt} is correlated with ξ_{jt} ; typically, ξ_{jt} will be correlated with p_{kt} .
- Also firms other than j can observe ξ_{jt} and respond strategically to this shock in their pricing decisions.

Endogeneity of Prices: Solutions

$$w_{jt} = \beta_j^{(0)} + \beta_j^{(y)} \ln(y_t) + \beta_{j1}^{(p)} \ln(p_{1t}) + \dots + \beta_{jJ}^{(p)} \ln(p_{Jt}) + \xi_{jt}$$

- We need to deal with the endogeneity problem due to $E(\ln(p_t)\xi_{jt}) \neq 0$.
- The most common approaches: (1) Instrumental variables; (2) Control function.
 - ① Instrumental variables:
 - (a) Input prices, costs
 - ② Control Function:
 - (a) Fixed effects
 - (b) Fixed effects - Cochrane-Orcutt

Table of Contents

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space**
- 6 Demand System in Characteristics Space

Some Limitations of Demand Systems in Product Space

- For the type of empirical questions in which we are interested in Empirical IO, demand systems in product space have several practical limitations.
 - ① Representative consumer assumption.
 - ② Too many parameters.
 - ③ Finding instruments for prices.
 - ④ Problems to predict demand of new varieties.

[1] Representative consumer assumption

- Very unrealistic. Propensity to substitute between different products is very heterogeneous across consumers.
- Ignoring this heterogeneity can generate substantial biases.
- In principle, the model can be applied to consumer/household level data. However:
 - Household-level data is often not available for some products/industries.
 - At the lower-stage, observed household choices seem discrete (only one variety) and this is at odds with this "continuous choice" model.

[2] Too many parameters

- The number of parameters is $2J + \frac{J(J+1)}{2}$, i.e., J intercept parameters (α); J income elasticities (γ); and $\frac{J(J+1)}{2}$ free price elasticities (β).
- It is not possible to estimate demand systems for differentiated products with many varieties.
- For instance, demand system for car models. With $J = 100$, the #parameters = 5,250.
- We need many thousands of observations (markets or/and time periods) to estimate this model. This type of data is typically not available.

[3] Finding instruments for prices

- Most applications of this class of models have ignored the potential endogeneity of prices.
- However, it is well known and simultaneity and endogeneity are potentially important issues in any demand estimation.
- The typical solution to this problem is using instrumental variables. In this model, the researcher needs at least as many instruments as prices, that is J .
- The ideal case is when we have information on production costs for each individual good. However, that information is very rarely available.

[4] Problems to predict demand of new varieties

- A problem that has received substantial attention is the prediction of the demand of a new product.
- Trajtenberg (1989), Hausman (1996), and Petrin (2002) are some prominent applications.
- In a demand system in product space, estimating the demand of a new good, say $J + 1$, requires estimates of the parameters associated with that good: $\beta_{J+1}^{(0)}$, $\beta_{J+1}^{(y)}$ and $\{\beta_{J+1,j} : j = 1, 2, \dots, J + 1\}$.
- This makes it impossible to make counterfactual predictions, i.e., predict the demand of a product that has not been introduced in any market yet.
- It also limits the applicability of this model in cases where the new product has been introduced very recently or in very few markets, because we may not have enough data to estimate these parameters.

Table of Contents

- 1 Motivation
- 2 Single product model
- 3 Basic Assumption
- 4 Demand systems in product space
- 5 Some Limitations of Demand Systems in Product Space
- 6 Demand System in Characteristics Space**

Basic Assumption

- The basic assumptions are:
 - ① A product, say a laptop computer, can be described in terms of a **bundle of physical characteristics**: e.g., CPU speed, memory, screen size, etc. These characteristics determine a product.
 - ② Consumers have **preferences on characteristics of products**, not on the products per se.
 - ③ A product category (e.g., laptops) has J different products and **each consumer buys at most one variety of the product per period**. This implies that the J products are substitutes with each other.

Model: Products

- We index varieties by $j \in \{1, 2, \dots, J\}$.
- We can distinguish two sets of product characteristics.
- **Characteristics observable to the researcher:**

$$X_j \equiv (X_{1j}, X_{2j}, \dots, X_{Kj})$$

X_{kj} represents the "amount" of attribute k in product j .

- Example (Laptops): X_{1j} = CPU speed; X_{2j} = RAM memory; X_{3j} = hard disk memory; X_{4j} = weight; X_{5j} = screen size; X_{6j} = dummy 'Intel inside'; etc.
- Other **characteristics are not observable to the researcher** but they are known and valued by consumers. We use ξ_j to represent the unobservable (to the researcher) characteristics of product j .

Model: Consumer Preferences

- The utility of household h if it purchases product j is:

$$U_{hj} = u(c_h) + X_j\beta_h + \xi_j + \epsilon_{hj}$$

- c_h is consumption of the "outside good" and $u(c_h)$ is the corresponding utility.
- $X_j\beta_h = \beta_{1h}X_{1j} + \beta_{2h}X_{2j} + \dots + \beta_{Kh}X_{Kj}$ is the utility from the observable characteristics.
- Coefficient β_{kh} represents the **marginal utility of this product characteristic** (e.g., RAM memory) for consumer h .
- ξ_j is utility from unobservable characteristics, the same for all consumers.
- ϵ_{hj} is an idiosyncratic taste of consumer h for product j that is unobservable to the researcher.

Model: Budget Constraint

- Budget constraint of consumer h , with income y_h , if buys product j :

$$c_h + p_j \leq y_h$$

- To represent no purchase of any of the J products we use $j = 0$, with $p_0 = 0$, $X_j = 0$, and $\xi_j = 0$.
- The utility of purchasing product j , after taking into account the budget constraints ($c_h = y_h - p_j$) becomes:

$$U_{hj} = u(y_h - p_j) + X_j\beta_h + \xi_j + \epsilon_{hj}$$

Model: Consumer Decision Problem

- A consumer chooses the alternative j from $\{0, 1, \dots, J\}$ that maximizes her utility U_{hj} .
- Consumer h chooses product j if and only if:

$$U_{hj} > U_{hi} \quad \text{for any } i \neq j$$

- Or equivalently, if and only if:

$$u(y_h - p_j) + X_j\beta_h + \xi_j + \epsilon_{hj} > u(y_h - p_i) + X_i\beta_h + \xi_i + \epsilon_{hi} \quad \text{for any } i \neq j$$

Model: Individual Consumer (Unit) Demands

- Let q_{hj} represent the quantity of product j demanded by consumer h .
- In this model, q_{hj} is either 0 or 1: $q_{hj} \in \{0, 1\}$.
- The demand is:

$$q_{hj} = \begin{cases} 1 & \text{if } U_{hj} > U_{hi} \text{ for any } i \neq j \\ 0 & \text{otherwise} \end{cases}$$

with $U_{hj} = u(y_h - p_j) + X_j\beta_h + \xi_j + \epsilon_{hj}$

- These unit demands depend on income, all the product prices, and all the product characteristics X_j and ξ_j .

Model: Aggregate Demand / Market Shares

- Let s_j be the proportion of consumers who purchase product j .

$$s_j = \frac{\sum_{h=1}^H q_{hj}}{H} = \Pr(q_{hj} = 1)$$

- We describe s_j as the **market share of product j** . According to the model:

$$\begin{aligned} s_j &= \Pr(U_{hj} > U_{hi} \text{ for any } i \neq j) \\ &= \Pr \left(\begin{array}{l} u(y_h - p_j) + X_j \beta_h + \xi_j + \epsilon_{hj} \\ > u(y_h - p_i) + X_i \beta_h + \xi_i + \epsilon_{hi} \\ \text{for any } i \neq j \end{array} \right) \end{aligned}$$

- This probability (market share) results from integrating the individual demand q_{hj} over the distribution of $(\beta_h, \epsilon_h, y_h)$.

General Model: Different Types of Discrete Choice Models

- The particular type of discrete choice model depends on three types of assumptions:
 - ① The specification of the "utility of money" $u(\cdot)$, e.g., linear or logarithmic function.
 - ② The probability distribution of the variables ϵ_{hj} , e.g., Logit or Probit.
 - ③ The assumptions on the coefficients β_h , e.g., constant ($\beta_h = \beta$ for every h) or heterogeneous coefficients.

General Model: Examples of Discrete Choice Models

- The typical three examples of discrete choice demand models that have received a lot of attention in empirical applications:
 - 1 Standard Logit model
 - Linear $u(\cdot)$; ϵ_{hj} i.i.d. Extreme Value; and $\beta_h = \beta$.
 - 2 Nested Logit model [not in this course!]
 - Linear $u(\cdot)$; ϵ_{hj} sum of two i.i.d. Extreme Values; and $\beta_h = \beta$.
 - 3 Random coefficients Logit model [not in this course!]
 - $u(\cdot)$ may be nonlinear; ϵ_{hj} i.i.d. Extreme Values; and β_h are heterogeneous across individuals according to some distribution.

Standard Logit Model: Extreme Value Distribution

- We say that a random variable ϵ has an Extreme Value type 1 distribution (or Gumbel distribution) if its Cumulative Distribution Function is:

$$F(\epsilon) = \exp(-\exp(-\epsilon))$$

- ϵ has support $(-\infty, +\infty)$. The density function is bell-shaped but asymmetric.
- Suppose that the variables $\epsilon_0, \epsilon_1, \dots, \epsilon_J$ are all Extreme Value and independently distributed. These variables satisfy two properties that are useful for discrete choice models.

Standard Logit Model: Extreme Value Distribution Properties [2]

- For any vector of constants $(\delta_0, \delta_1, \dots, \delta_J)$, we have that:

$$\Pr(\delta_j + \epsilon_j > \delta_i + \epsilon_i \text{ for any } i \neq j) = \frac{\exp(\delta_j)}{\sum_{i=0}^J \exp(\delta_i)}$$

- And:

$$E \left[\max_{j \in \{0,1,\dots,J\}} (\delta_j + \epsilon_j) \right] = \log \left(\sum_{j=0}^J \exp(\delta_j) \right)$$

Standard Logit Model: Demand Model

- Suppose that $u(c) = \alpha c$ [Linear]; $\beta_h = \beta$ for every h ; and all the ϵ_j 's are i.i.d. Extreme Value Type 1.
- Then, the utility of household h if it purchases product j is:

$$U_{hj} = \alpha p_j + X_j \beta + \xi_j + \epsilon_{hj}$$

- If we define $\delta_j = \alpha p_j + X_j \beta + \xi_j$, then δ_j represents the average utility of purchasing product j for every consumer in the market.
- The only idiosyncratic component of utility is ϵ_{hj} .
- The model implies that the market shares are:

$$s_j = \frac{\exp(\delta_j)}{1 + \sum_{i=1}^J \exp(\delta_i)} = \frac{\exp(-\alpha p_j + X_j \beta + \xi_j)}{1 + \sum_{i=1}^J \exp(-\alpha p_i + X_i \beta + \xi_i)}$$

because $\delta_0 = -\alpha p_0 + X_0 \beta + \xi_0 = 0$.

Standard Logit Model: Parameters

- When the indirect utility has this form,

$$U_{hj} = -\alpha p_j + \beta_1 X_{1j} + \dots + \beta_K X_{Kj} + \xi_j + \epsilon_{hj}$$

- We have that the parameters β and α have a very clear economic interpretation:
 - α is the marginal utility of income (or marginal utility of money), and it is measured in "utils" per dollar.
 - β_k is the marginal utility of attribute k , and it is measured in "utils" per unit of this attribute.
 - Therefore, $\frac{\beta_k}{\alpha} = \frac{\text{utils/unit of } k}{\text{utils/\$}} = \frac{\$}{\text{unit of } k}$ is the *marginal value* of attribute k measured in dollars per unit of the attribute.

Standard Logit Model: Elasticities

- In the logit model, $s_j = \frac{\exp(\delta_j)}{1 + \sum_{i=1}^J \exp(\delta_i)}$, implies that:

$$\frac{ds_j}{d\delta_j} = s_j(1 - s_j) \quad \text{and} \quad \frac{ds_j}{d\delta_i} = -s_j s_i \text{ for } i \neq j$$

- Therefore, the own price elasticities are:

$$\frac{ds_j}{dp_j} \frac{p_j}{s_j} = \left[\frac{ds_j}{d\delta_j} \frac{d\delta_j}{dp_j} \right] \frac{p_j}{s_j} = s_j(1 - s_j)[- \alpha] \frac{p_j}{s_j} = -\alpha(1 - s_j)p_j$$

- And the cross price elasticities are:

$$\frac{ds_j}{dp_i} \frac{p_i}{s_j} = \left[\frac{ds_j}{d\delta_i} \frac{d\delta_i}{dp_i} \right] \frac{p_i}{s_j} = -s_j s_i [- \alpha] \frac{p_i}{s_j} = \alpha s_i p_i$$

Standard Logit Model: Estimation

- According to the model, for any product $j = 1, 2, \dots, J$:

$$\frac{s_j}{s_0} = \exp(-\alpha p_j + X_j \beta + \xi_j)$$

- Thus:

$$\log(s_j) - \log(s_0) = -\alpha p_j + X_j \beta + \xi_j$$

- This is a linear regression model with regressors (p_j, X_j) , parameters (α, β) , and error term ξ_j .
- If the number of product J is large enough relative to the number of observable product characteristics K , then we can estimate the parameters of this model using a cross-section of consumers (one market).

Standard Logit Model: Some Advantages

- Let's revisit the limitations of traditional demand systems and see how the Logit model overcomes these limitations:
 - 1 Positive consumption of all products. As a discrete choice model, the logit does not have this unrealistic prediction.
 - 2 "Too many" parameters. The number of parameters (α, β) is $K + 1$ that does not increase with the number of products J . Having many products (large J) is a blessing for this model (more degrees of freedom, more precise estimates).
 - 3 Finding instruments for prices. There is only one endogenous regressor in the model. We need only one instrument.
 - 4 Predict demand of new varieties. Given (α, β) , we can predict the demand of any hypothetical product with characteristics (p, X) .

Standard Logit Model: Endogeneity

- We have the regression model:

$$\log(s_j) - \log(s_0) = \beta^{(p)} p_j + X_j \beta + \xi_j$$

- Endogeneity of prices: Products with better (higher) unobserved quality ξ_j tend to have higher prices: $\text{cov}(p_j, \xi_j) > 0$.
- This endogeneity problem implies that the OLS estimate of the parameter associated to price is upward biased:

$$\hat{\beta}(p) \rightarrow \beta(p) + \frac{\text{cov}(p_j, \xi_j)}{\text{var}(p_j)}$$

- We could even get $\hat{\beta}(p)$ is positive or not significantly different from zero.

Standard Logit Model: IV Estimation

- BLP instruments: Berry, Levinsohn, & Pakes (1996) proposed the following instrument for price in discrete choice models.
- Suppose that price is the only endogenous regressor such that $E[X_j \xi_j] = 0$.
- Firms' price competition implies that the price of a product depends on its characteristics (X_j, ξ_j) , but also on the characteristics of the competing products $(X_i, \xi_i : i \neq j)$.
- Intuitively, if the other products have characteristics very similar to j , competition is more intense and p_j will be smaller. But if product j has very different characteristics from the other products, firm j can enjoy some market power from product differentiation and p_j can be larger.
- The idea is using $(X_i : i \neq j)$ as an instrument for p_j .

Standard Logit Model: IV Estimation - BLP Instruments [2]

- Suppose that we use the vector of instruments $Z_j = (Z_{1j}, Z_{2j}, \dots, Z_{Kj})$ where for any product characteristic k ,

$$Z_{kj} = \frac{\sum_{i \neq j} X_{ki}}{J - 1}$$

- We could consider other functions of $(X_i : i \neq j)$ as instruments, such as the average distance,

$$Z_{kj} = \frac{\sum_{i \neq j} |X_{ki} - X_{kj}|}{J - 1}$$

Standard Logit Model: IV Estimation - BLP Instruments [3]

- Z_j is a vector of valid instruments for p_j in the regression equation:

$$\log(s_j) - \log(s_0) = \beta^{(p)} p_j + X_j \beta + \xi_j$$

- 1 They are not regressors in this regression function (the utility of product j depends only on characteristics of product j);
- 2 They are not correlated with the error term ξ_j (by assumption);
- 3 They are correlated with p_j (testable restriction).

Estimating Production Functions

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- This presentation refers to chapter 3 (3.1, 3.2, 3.3.1, 3.4.1, 3.4.2, 3.4.3) of **Empirical Industrial Organization: Models, Methods, and Applications** by Victor Aguirregabiria.

Outline

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

What is a Production Function (PF)?

It is a function that **relates** the **amount of physical output** of a production process (Y) to the **amount of physical inputs** or factors of production (X). Estimation of PFs plays a key role in empirical questions such as:

- Estimation of Firms' Costs.
- Measurement of Firms' Productivity (and its growth).
- Misallocation of inputs. Do more productive firms use more inputs?
- Evaluating the effects of adopting new technologies.
- Measurement of Learning-by-doing.

Production Functions

- A standard representation of a PF is:

$$Y = A \times f(X_1, X_2, \dots, X_J)$$

- Y is the firm's output (physical units per year).
- $X_1, X_2, \dots, X_J$ are measures of J firm inputs.
Labor, capital, energy, materials, ...
- A represents the firm's Total Factor Productivity.
Anything else affecting output that we do not observe as researchers:
managerial ability, organization of production, quality of inputs,
quality of land, ...
- The marginal productivity of input j is: $MP_j = \frac{dY}{dX_j} = A \frac{df}{dX_j}$.
- Note that TFP increases proportionally the MP of all the inputs.

Cobb-Douglas Production Function

- A common specification is the Cobb-Douglas PF:

$$Y = AX_1^{\alpha_1} X_2^{\alpha_2} \dots X_J^{\alpha_J}$$

$\alpha_1, \alpha_2, \dots, \alpha_J$ are technological parameters (all positive).

- For the Cobb-Douglas PF the marginal productivity of input j is:

$$MP_j = \frac{dY}{dX_j} = \alpha_j AX_1^{\alpha_1} \dots X_j^{\alpha_j-1} \dots X_J^{\alpha_J} = \alpha_j \frac{Y}{X_j}$$

- In this PF, MP_j increases with the amount of any other input $k \neq j$. This means that **all the inputs are complements in production**.

$$\frac{dMP_j}{dX_k} = \frac{\alpha_j}{X_j} \frac{dY}{dX_k} = \frac{\alpha_j \alpha_k Y}{X_j X_k} > 0$$

Production Function and Cost Function

- Given a firm's PF and input prices, its **cost function** $C(Y)$ is defined as the minimum cost of producing the amount of output Y :

$$C(Y) = \left[\begin{array}{l} \min_{\{X_1, X_2, \dots, X_J\}} W_1 X_1 + W_2 X_2 + \dots + W_J X_J \\ \text{subject to: } Y = Af(X_1, X_2, \dots, X_J) \end{array} \right]$$

where $W_1, W_2, \dots, W_J$ are input prices.

- Or using a Lagrange representation:

$$C(Y) = \min_{\{\lambda, X_1, \dots, X_J\}} W_1 X_1 + \dots + W_J X_J + \lambda[Y - Af(X_1, \dots, X_J)]$$

where λ is the Lagrange multiplier of the output restriction.

- The **marginal conditions of optimality** imply that for every input j :

$$W_j - \lambda MP_j = 0$$

Cost Function for the Cobb-Douglas PF

- Remember that for the Cobb-Douglas PF $MP_j = \alpha_j \frac{Y}{X_j}$.
- Therefore, the marginal condition of optimality for input j implies:

$$W_j X_j = \lambda \alpha_j Y$$

And the total cost is equal to:

$$\sum_{j=1}^J W_j X_j = \lambda \alpha Y, \text{ where } \alpha \equiv \sum_j \alpha_j$$

- Parameter α measures the **Returns to Scale** in the PF.
- Returns to Scale: constant (CRS) if $\alpha = 1$, decreasing (DRS) if $\alpha < 1$, and increasing (IRS) if $\alpha > 1$.

Derivation of Cost Function for Cobb-Douglas [2]

- We need to obtain the value of the Lagrange multiplier λ . For this, we solve the marginal conditions $X_j = \lambda \alpha_j Y / W_j$ into the PF:

$$Y = A \left(\frac{\lambda \alpha_1 Y}{W_1} \right)^{\alpha_1} \left(\frac{\lambda \alpha_2 Y}{W_2} \right)^{\alpha_2} \cdots \left(\frac{\lambda \alpha_J Y}{W_J} \right)^{\alpha_J}$$

- Solving in this expression for the Lagrange multiplier:

$$\lambda = \left(\frac{W_1}{\alpha_1} \right)^{\frac{\alpha_1}{\alpha}} \left(\frac{W_2}{\alpha_2} \right)^{\frac{\alpha_2}{\alpha}} \cdots \left(\frac{W_J}{\alpha_J} \right)^{\frac{\alpha_J}{\alpha}} \left(\frac{Y}{A} \right)^{\frac{1}{\alpha}} \frac{1}{Y}$$

- Plugging this expression of the multiplier into the equation $C(Y) = \lambda \alpha Y$ for the cost, we obtain the cost function:

$$C(Y) = \alpha \left(\frac{Y}{A} \right)^{\frac{1}{\alpha}} \left(\frac{W_1}{\alpha_1} \right)^{\frac{\alpha_1}{\alpha}} \left(\frac{W_2}{\alpha_2} \right)^{\frac{\alpha_2}{\alpha}} \cdots \left(\frac{W_J}{\alpha_J} \right)^{\frac{\alpha_J}{\alpha}}$$

Cost Function for Cobb-Douglas [3]

$$C(Y) = \alpha \left(\frac{Y}{A} \right)^{\frac{1}{\alpha}} \left(\frac{W_1}{\alpha_1} \right)^{\frac{\alpha_1}{\alpha}} \left(\frac{W_2}{\alpha_2} \right)^{\frac{\alpha_2}{\alpha}} \dots \left(\frac{W_J}{\alpha_J} \right)^{\frac{\alpha_J}{\alpha}}$$

- The marginal cost is:

$$C'(Y) = Y^{\frac{1}{\alpha}-1} \left(\frac{1}{A} \right)^{\frac{1}{\alpha}} \left(\frac{W_1}{\alpha_1} \right)^{\frac{\alpha_1}{\alpha}} \left(\frac{W_2}{\alpha_2} \right)^{\frac{\alpha_2}{\alpha}} \dots \left(\frac{W_J}{\alpha_J} \right)^{\frac{\alpha_J}{\alpha}}$$

- The sign of $C''(Y)$ is equal to the sign of $\frac{1}{\alpha} - 1$.
 - $\alpha = 1$ (constant returns): $C''(Y) = 0$ (Constant MC)
 - $\alpha < 1$ (decreasing returns): $C''(Y) > 0$ (Increasing MC)
 - $\alpha > 1$ (increasing returns): $C''(Y) < 0$ (Decreasing MC).

More on the Cobb-Douglas

- A nice property of the Cobb-Douglas (for the purpose of estimation) is that its **logarithm transformation is linear in parameters**:

$$\ln(Y) = \log(A) + \alpha_1 \log(X_1) + \alpha_2 \log(X_2) + \dots + \alpha_J \log(X_J)$$

- We will represent $\log(Y)$ and $\log(X)$ using the lower letters y and x , resp., and the log-TFP using ω , such that:

$$y = \omega + \alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_J x_J$$

- Differences in log-TFP (in ω) are in percentage terms:
Example: Consider two firms, 1 and 2, using the same amount of inputs X but with $\omega_1 = 1.1$ and $\omega_2 = 1.5$ such that $\omega_2 - \omega_1 = 0.4$. Therefore, firm 2 is 40% more productive than firm 1.

More on the Cobb-Douglas [2]

- Most empirical applications that we will see in this course consider two inputs: **Labor** (L) and **Capital** (K):

$$y = \alpha_L \ell + \alpha_K k + \omega$$

with $\ell \equiv \log(L)$ and $k \equiv \log(K)$. Sometimes the specification also includes **Materials** (M):

$$y = \alpha_L \ell + \alpha_K k + \alpha_M m + \omega$$

with $m \equiv \log(M)$.

Table of Contents

- 1 Model
- 2 Data**
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Data

- Panel Data of N firms over T time periods with information on output, labor, and capital (in logs):

$$\{y_{it}, \ell_{it}, k_{it} : i = 1, 2, \dots, N; t = 1, 2, \dots, T\}$$

We are interested in the estimation of the parameters α_L and α_K in the Cobb-Douglas PF (in logs):

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \omega_{it} + e_{it}$$

ω_{it} = log-TFP. Unobserved inputs (for the researcher) which are known to the firm when it decides K and L : e.g., managerial ability, quality of land, different technologies.

Observing Revenue instead of Physical Output

- Revenue = $R_{it} = P_{it} Y_{it}$ such that:

$$\log(R_{it}) = \log(P_{it}) + \log(Y_{it}) = p_{it} + y_{it}$$

but the researcher only observes $\log(R_{it})$.

- **Solution 1**

- Proxy $\log(P_{it})$ using industry-level price index, $\log(P_{Industry,t})$.
- $\log(P_{it}) = \log(P_{Industry,t}) + u_{it}$, where u_{it} is measurement error.
- This measurement error can be interpreted as part of the TFP ω_{it} .

Observing Revenue instead of Physical Output [2]

- **Solution 2** Assume isoelastic demand & monopolistic competition:

$$y_{it} = b_{it} - \beta p_{it}$$

b_{it} is unobservable to the researcher and β is the demand elasticity.

- Then, $p_{it} = \frac{b_{it} - y_{it}}{\beta}$ and:

$$\log(R_{it}) = p_{it} + y_{it} = \frac{b_{it}}{\beta} + \left(1 - \frac{1}{\beta}\right) y_{it}$$

such that:

$$\log(R_{it}) = \alpha_L^* \ell_{it} + \alpha_K^* k_{it} + \omega_{it}^* + e_{it}$$

with $\alpha_L^* = \left(1 - \frac{1}{\beta}\right) \alpha_L$, $\alpha_K^* = \left(1 - \frac{1}{\beta}\right) \alpha_K$

- This is very relevant for interpretation of results and of "log-TFP".
- For instance, $\alpha_L^* + \alpha_K^* = \left(1 - \frac{1}{\beta}\right) (\alpha_L + \alpha_K) < \alpha_L + \alpha_K$

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?**
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Total Factor Productivity (TFP)

- Production function:

$$Y_{it} = A_{it}F(K_{it}, L_{it}, M_{it})$$

A_{it} is denoted Total Factor Productivity (TFP).

- It is a factor-neutral shifter that captures variations in output not explained by observable inputs.
- TFP is a residual.

Large & Persistent Differences in TFP across Firms

- **Ubiquitous:** even within narrowly defined industries and products.
- **Large:** 90th to 10th percentile TFP ratios: $\frac{A_{90th}}{A_{10th}}$
 - U.S. manufacturing, **average** within 4-digit SIC industries = **1.92**
 - Denmark: average = **3.75**
 - China or India, average > 5
- **Persistent:** AR(1) of log-TFP with annual frequency: autoregressive coefficients between 0.6 to 0.8.
- **It matters:** Higher TFP producers are more likely to survive, innovate, invest, ...

Why do Firms differ in their Productivity Levels?

- What supports such large productivity differences in equilibrium?
- Can producers control the factors that influence productivity or are they purely external effects of the environment?
- If firms can partly control their TFP, what type of choices increase it?

Why TFP dispersion is Possible in Equilibrium?

- Because the profit function is concave in output and the optimal amount of profit for a monopolist (or duopolist, ...) is smaller than total demand.
- Let the profit of a firm be:

$$\pi_i = P_i(Y_i)Y_i - C(Y_i, A_i)$$

$P_i(Y_i)$ = Inverse demand function; $C(Y_i, A_i)$ = Cost function.

- **Key condition:** either $P_i(Y_i)Y_i$ is strictly concave in Y_i , or $C(.)$ is strictly convex in Y_i . [The profit function is strictly concave].
- **Example: DRS even with perfect competition.** PY_i is linear in Y_i but $C(.)$ is strictly convex because DRS.
- **Example: Oligopoly competition even with CRS.** $C(.)$ is linear but $P_i(Y_i)Y_i$ is strictly concave in Y_i if demand is downward sloping

Why TFP Dispersion is Possible in Equilibrium? [2]

- Equilibrium implies the marginal condition for optimal output:

$$MR_i \equiv \frac{\partial [P(Y_i)Y_i]}{\partial Y_i} = \frac{\partial C(Y_i, A_i)}{\partial Y_i} \equiv MC_i$$

- If variable profit is strictly concave, this equilibrium can support firms with different TFPs, A_i .
- It is not optimal for the firm with highest TFP to provide all the output in the industry.
- Firms with different TFPs (above a certain threshold value) operate in the same market.

How can a firm affect its TFP?

- Human resources and Managerial Practices.
- Learning-by-Doing
- Organizational structure: vertical integration vs outsourcing.
- Higher-Quality of Labor and Capital inputs.
- Adoption of new technologies.
- Investment in R&D.
- Innovation: process and product innovation.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem**
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

The Simultaneity Problem

- Consider the PF:

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \omega_{it}$$

- We are interested in the estimation of α_L and α_K .
- These parameters represent "ceteris paribus" causal effects of labor and capital on output, respectively.
- When the manager decides the optimal (k_{it}, ℓ_{it}) she has some information about log-TFP ω_{it} (that we do not observe).
- This means that there is a correlation between the observable inputs (k_{it}, ℓ_{it}) and the unobservable ω_{it} .
- This correlation implies that the OLS estimators of α_L and α_K are biased and inconsistent.

Simultaneity Problem

- Consider the Cobb-Douglas PF in logarithms:

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \omega_{it} + e_{it}$$

- We want to estimate parameters α_L and α_K . These parameters represent the causal effects of labor and capital on output.
- When the manager decides the optimal (k_{it}, ℓ_{it}) , she has some information about log-TFP ω_{it} .
- This means that there is a correlation between the observable inputs (k_{it}, ℓ_{it}) and the unobservable ω_{it} .
- This correlation implies that the OLS estimates of α_L and α_K are biased and inconsistent.

Simultaneity Problem: How do we know?

- How do we know whether $E(x_i\epsilon_i) = 0$ or $E(x_i\epsilon_i) \neq 0$?
- In general, we don't know, but in many cases we can have serious suspicion of omitted variables that are correlated with the regressor(s).
- Only when the observable regressor comes from a randomized experiment we can be certain that $E(x_i\epsilon_i) = 0$.
- But data from randomized experiments are still rare in many applications in economics.

Simultaneity Problem: How do we know? [2]

- In models with **simultaneous equations**, the model itself can tell us that some regressors are correlated with the error term: $E(x_i \epsilon_i) \neq 0$.
- For instance, this is the case in the production function model once we take into account the firm's optimal demand for inputs.

Simultaneity Problem: Example [1/3]

- A Cobb-Douglas PF only with labor input:

$$Y_i = A_i L_i^{\alpha_L}$$

- The amounts of output (Y_i) and labor (L_i) are endogenous variables which are determined by the conditions of profit maximization.
- Firms operate in the same markets for output and inputs. Same output and input prices: P and W .
- A firm's profit is: $\pi_i = PY_i - WL_i$
- A firm's Labor Demand is the amount L_i that maximizes profit:

$$\frac{d\pi_i}{dL_i} = 0 \rightarrow MP_{L_i} = \frac{W}{P} \rightarrow \alpha_L \frac{Y_i}{L_i} = \frac{W}{P}$$

Simultaneity Problem: Example [2/3]

- The complete model consists of the Production Function (PF) and the Labor Demand equation (LD):

$$(PF) \quad Y_i = A_i L_i^{\alpha_L}$$

$$(LD) \quad L_i = \frac{\alpha_L Y_i}{W/P}$$

- This is a system of two equations with two endogenous variables.
- We can take logarithms in these equations to have a model that is linear in parameters (a linear regression model):

$$(\log\text{-PF}) \quad y_i = \alpha_0 + \alpha_L \ell_i + \omega_i$$

$$(\log\text{-LD}) \quad \ell_i = \gamma_0 + y_i$$

where α_0 is the mean value of $\log(A_i)$; $\omega_i = \log(A_i) - \alpha_0$; and $\gamma_0 = \log(\alpha_L P/W)$.

Simultaneity Problem: Example [3/3]

- Solving for the endogenous variables in the **system of equations**,

$$(\text{log-PF}) \quad y_i = \alpha_0 + \alpha_L \ell_i + \omega_i$$

$$(\text{log-LD}) \quad \ell_i = \gamma_0 + y_i$$

- we obtain the solution:

$$y_i = \frac{\omega_i + \alpha_0 + \alpha_L \gamma_0}{1 - \alpha_L}$$

$$\ell_i = \frac{\omega_i + \alpha_0 + \gamma_0}{1 - \alpha_L}$$

- This solution shows that ℓ_i is correlated with ω_i :

$$\text{Cov}(\ell_i, \omega_i) = \frac{\text{Var}(\omega_i)}{(1 - \alpha_L)^2} > 0$$

Simultaneity Problem: Biased OLS

- Following up with this example, we can show that the OLS estimator of α_L is biased.

$$\hat{\alpha}_L^{\text{OLS}} = \frac{S_{\ell y}}{S_{\ell \ell}} = \frac{\sum_{i=1}^N (y_i - \bar{y})(\ell_i - \bar{\ell})}{\sum_{i=1}^N (\ell_i - \bar{\ell})^2}$$

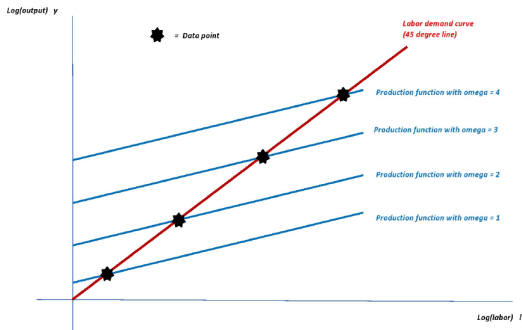
- The model implies that:

$$y_i - \bar{y} = \frac{\omega_i}{1 - \alpha_L}$$

$$\ell_i - \bar{\ell} = \frac{\omega_i}{1 - \alpha_L}$$

Such that $\hat{\alpha}_L^{\text{OLS}} = \frac{S_{\ell y}}{S_{\ell \ell}} = 1$, and $\text{Bias (OLS)} = 1 - \alpha_L$.

Simultaneity Problem: Graphical Representation



- Graphical representation of structural equations in space (ℓ, y) .
- Graphical interpretation of the bias of the OLS estimator.
- With sample variation in the log-real-wage w_i , the bias will be reduced, but it will be always present.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions**
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Solutions to the Simultaneity Problem

- We are going to consider two possible solutions to the endogeneity problem:
 - ① Control function / Fixed effects estimation
 - ② Instrumental variables estimation.
- First, we will see these potential solutions in a general regression model, and then we will particularize them to the estimation of PFs.

Control Function Method

- Consider the LRM:

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_K x_{Ki} + \epsilon_i$$

where we are concerned about the endogeneity of regressor x_{1i} , i.e., $E(x_{1i}\epsilon_i) \neq 0$.

- Suppose that the researcher has sample data for a variable c_i ("the control") that satisfies **two conditions**.
- [Control]** $\epsilon_i = \gamma c_i + u_i$ such that u_i is independent of x_{1i} and c_i .
- [No multicollinearity]** We cannot write c_i as a linear combination of the exogenous regressors $x_{2i}, \dots, x_{Ki}$.
- Under these conditions we can construct a consistent estimator of $\beta_1, \beta_2, \dots, \beta_K$: the Control Function (CF) estimator.

Control Function Estimator

- To obtain the CF estimator we simply include the CF variable c_i in the regression and apply OLS:

$$y_i = \beta_0 + \beta_1 x_{1i} + \dots + \beta_K x_{Ki} + \gamma c_i + u_i$$

- Under the "Control" condition, the new error term u_i is not correlated with the regressors.
- And under the "No multicollinearity" condition all the regressors (including c_i) are not linearly independent.
- Therefore, this OLS estimator is consistent.
- The CF approach uses observables to control for the part of the error that is correlated with the regressor.

Simultaneity Problem: Instrumental Variables

- Consider the LRM:

$$y_i = \beta_1 x_{1i} + \dots + \beta_K x_{Ki} + \epsilon_i$$

where we are concerned about the endogeneity of regressor x_{1i} , i.e., $E(x_{1i}\epsilon_i) \neq 0$.

- Suppose that the researcher has sample data for a variable z_i ("the instrument") that satisfies **two conditions**.
- [Relevance]** In a regression of x_{1i} on $(z_i, x_{2i}, \dots, x_{Ki})$, regressor z_i has a significant effect on x_{1i} .
- [Independence]** z_i is NOT correlated with ϵ_i : $E(z_i\epsilon_i) = 0$.
- Under these conditions we can construct a consistent estimator of $\beta_1, \beta_2, \dots, \beta_K$: the IV or Two-stage Least Square (2SLS) estimator.

Two Stage Least Squares (2SLS or IV Estimator)

- The IV or 2SLS can be implemented as follows.
- **[Stage 1]** Run an OLS regression of x_{1i} on $(z_i, x_{2i}, \dots, x_{Ki})$. Obtain the fitted values from this regression:

$$\hat{x}_{b1i} = \hat{\gamma}_0 + \hat{\gamma}_1 z_i + \hat{\gamma}_2 x_{2i} + \dots + \hat{\gamma}_K x_{Ki}$$

- **[Stage 2]** Run an OLS regression of y_i on $(\hat{x}_{b1i}, x_{2i}, \dots, x_{Ki})$. This OLS estimator is consistent for $\beta_1, \beta_2, \dots, \beta_K$.
- The first stage decomposes x_{1i} in two parts: $x_{1i} = \hat{x}_{1i} + e_{1i}$, where e_{1i} is the residual from this first-stage regression.
- Since $\hat{x}_{1i}$ depends only on exogenous regressors, it is not correlated with ϵ_i .

How do we find instruments?

- A simultaneous equation model may suggest valid instruments.
- For instance, consider the PF with only labor input, but now firms operate in different output/labor markets with different prices.

$$(PF) \ y_i = \alpha \ell_i + \omega_i$$

$$(LD) \ \ell_i = \log(\alpha) + y_i - w_i$$

with $w_i = \log(W_i/P_i)$.

- Suppose that the researcher observes w_i .
- It is clear that w_i satisfies the relevance condition: it does not enter in the PF as a regressor; it has an effect on labor.
- Under the condition $E(w_i \omega_i) = 0$ it is a valid instrument.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs**
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Input prices as Instruments (an IV estimator)

- If input prices r_{it} are observable (wages, cost of capital, fuel and energy prices), then under the assumption that they are not correlated with TFP, $E(\omega_{it}r_{it}) = 0$, we can use them as instruments.
- This approach has several **limitations/problems**.
- **Problem (1)**: Firms in the same industry typically use very similar type of inputs (labor, capital equipment, energy, materials) and they buy these inputs in the same input markets. If these input markets are competitive, the input prices are the same for all the firms in the industry:

$$r_{it} = r_t \quad \text{for every firm } i$$

- If input prices vary only over time, they are perfectly collinear with time-dummies in the PF. No valid instruments.

Input prices as IVs [2]

- **Problem (2):** When input prices have cross-sectional variation, it could be because endogenous reasons.
- (a) Inputs markets are not competitive and firms with higher productivity pay higher prices. Then, $\text{cov}(\omega_{it}, r_{it}) \neq 0$, making input prices not a valid instrument.
- (b) Firms may be using different types of labor or capital inputs, with different qualities. This difference in the quality of inputs is part of the log-TFP. Then, $\text{cov}(\omega_{it}, r_{it}) \neq 0$, making input prices not a valid instrument.

Input prices as IVs [3]

- An ideal situation for using input prices as IVs is when firms in the same industry produce in different geographic markets where the input markets are competitive.
- The variation in input prices over geographic markets is due to different conditions on the supply of inputs (e.g., labor supply, better access to materials) and not to differences in productivity.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)**
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Fixed-Effects (FE) Estimator

- Consider the PF:

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \omega_{it} \quad (1)$$

Let's first define the FE (or Within-Groups) estimator and then we will show under which conditions this estimator provides unbiased (consistent) estimates of parameters α_L and α_K .

- If, for each firm i , we average equation (1) considering all the years observations, we have the equation:

$$\bar{y}_i = \alpha_L \bar{\ell}_i + \alpha_K \bar{k}_i + \bar{\omega}_i \quad (2)$$

where:

$$\bar{y}_i = \frac{1}{T} \sum_{t=1}^T y_{it}, \quad \bar{\ell}_i = \frac{1}{T} \sum_{t=1}^T \ell_{it}, \quad \bar{k}_i = \frac{1}{T} \sum_{t=1}^T k_{it}, \quad \bar{\omega}_i = \frac{1}{T} \sum_{t=1}^T \omega_{it}$$

FE estimator [2]

- If we subtract equation (2) from equation (1), we have:

$$(y_{it} - \bar{y}_i) = \alpha_L(\ell_{it} - \bar{\ell}_i) + \alpha_K(k_{it} - \bar{k}_i) + (\omega_{it} - \bar{\omega}_i)$$

- This equation is named the **Fixed-Effects** (or the Within-Groups) **transformation of the model**.
- The **FE estimator** is OLS applied to the FE transformed model.
- For instance, if we had only one input, say labor, the FE estimator of α_L would be:

$$\widehat{\alpha_L^{\text{OLS}}} = \frac{\sum_{i=1}^N \sum_{t=1}^T (y_{it} - \bar{y}_i)(\ell_{it} - \bar{\ell}_i)}{\sum_{i=1}^N \sum_{t=1}^T (\ell_{it} - \bar{\ell}_i)^2}$$

FE estimator in Stata: Implementation 1

- Let `logy`, `logn`, `logk`, `id`, `year`, be the variables.
- First, we construct the within-group transformation of the variables:

```
egen mlogy = mean(logy), by(id)
gen wlogy = logy - mlogy
egen mlogn = mean(logn), by(id)
gen wlogn = logn - mlogn
egen mlogk = mean(logk), by(id)
gen wlogk = logk - mlogk
```

Implementation 1

Then, run OLS using within-groups transformed variables:

```
reg wlogy wlogn wlogk
```

Source	SS	df	MS	Number of obs	=	4,072
				F(2, 4069)	=	5255.44
Model	211.639421	2	105.81971	Prob > F	=	0.0000
Residual	81.9304197	4,069	.020135271	R-squared	=	0.7209
				Adj R-squared	=	0.7208
Total	293.56984	4,071	.072112464	Root MSE	=	.1419

wlogy	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
wlogn	.5995702	.012516	47.90	0.000	.575032	.6241084
wlogk	.3190445	.0092708	34.41	0.000	.3008687	.3372203
_cons	4.53e-09	.0022237	0.00	1.000	-.0043597	.0043597

FE estimator: Implementation 2

Define i and t in panel data. Then, we apply FE estimator:

```
xtset id year
```

```
xtreg logy logn logk, fe
```

```
Fixed-effects (within) regression
Group variable: id
```

```
R-sq:
```

```
  within = 0.7209
  between = 0.9729
  overall = 0.9683
```

```
corr(u_i, Xb) = 0.4651
```

```
Number of obs   =    4,072
Number of groups =     509
```

```
Obs per group:
```

```
   min =      8
   avg =     8.0
   max =      8
```

```
F(2,3561)       =   4599.32
Prob > F        =    0.0000
```

logy	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
logn	.5995702	.013379	44.81	0.000	.5733389	.6258015
logk	.3190445	.00991	32.19	0.000	.2996146	.3384744
_cons	3.512405	.0404574	86.82	0.000	3.433083	3.591727
sigma_u	.37142884					
sigma_e	.15168289					
rho	.85706581	(fraction of variance due to u_i)				

```
F test that all u_i=0: F(508, 3561) = 36.48
```

```
Prob > F = 0.0000
```

FE estimator: Implementation 2

We can also include year dummies:

`xtreg logy logn logk i.year, fe`

Fixed-effects (within) regression
Group variable: id

R-sq:

within = 0.7379
between = 0.9706
overall = 0.9661

`corr(u_i, Xb)` = 0.5988

Number of obs = 4,072
Number of groups = 509

Obs per group:

min = 8
avg = 8.0
max = 8

F(9, 3554) = 1111.47
Prob > F = 0.0000

logy	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
logn	.6544609	.0144048	45.43	0.000	.6262184	.6827034
logk	.2329072	.013637	17.08	0.000	.2061702	.2596443
year						
1983	-.0376406	.0093042	-4.05	0.000	-.0558828	-.0193985
1984	-.0076445	.0096071	-0.80	0.426	-.0264805	.0111914
1985	-.0234513	.0100955	-2.32	0.020	-.0432449	-.0036578
1986	-.0136103	.0105543	-1.29	0.197	-.0343034	.0070829
1987	.0314121	.0108748	2.89	0.004	.0100907	.0527335
1988	.0753576	.0111072	6.78	0.000	.0535805	.0971347
1989	.0764164	.0118166	6.47	0.000	.0532485	.0995844

Consistency of FE estimator

- The FE estimator is OLS in the regression equation:

$$(y_{it} - \bar{y}_i) = \alpha_L(\ell_{it} - \bar{\ell}_i) + \alpha_K(k_{it} - \bar{k}_i) + (\omega_{it} - \bar{\omega}_i)$$

- As any OLS estimator, it is consistent if the error term is not correlated with the regressors. In this case, this implies:

$$E[(\omega_{it} - \bar{\omega}_i)(\ell_{it} - \bar{\ell}_i)] = E[(\omega_{it} - \bar{\omega}_i)(k_{it} - \bar{k}_i)] = 0$$

- We now present two **assumptions** on the unobserved log-TFP that imply consistency of the FE estimator (with time dummies).

Consistency of FE estimator [2]

- **Assumption FE-1.** Log-TFP has the following structure:

$$\omega_{it} = \eta_i + \delta_t + u_{it}$$

- η_i is interpreted as managerial ability, or a different technology that is constant over time.
- δ_t represents productivity that affects in the same way to all the firms in the industry.
- u_{it} is a firm-specific transitory shock.

Consistency of FE estimator [3]

- **Assumption FE-2.** The firm-specific transitory shock, u_{it} , is not correlated over time and it is realized after the firm chooses the amount of inputs at period t .
- u_{it} is a "surprise" that is realized after the firm has chosen inputs. For any two periods t and s , u_{it} is not correlated with inputs ℓ_{is} and k_{is} .

Consistency of FE estimator [4]

- Under assumptions FE-1 we have that:

$$\bar{\omega}_i = \eta_i + \bar{\delta} + u_i$$

such that:

$$\omega_{it} - \bar{\omega}_i = \delta_t - \bar{\delta} + u_{it} - \bar{u}_i.$$

- If we control for $\delta_t - \bar{\delta}$ using time dummies, the remaining error term is $u_{it} - \bar{u}_i$.
- Under Assumption FE-2, the error term $u_{it} - \bar{u}_i$ is not correlated with the regressors $(\ell_{it} - \bar{\ell}_i)$ and $(k_{it} - \bar{k}_i)$ because, for any two periods t and s , u_{it} is not correlated with inputs ℓ_{is} and k_{is} .
- Under FE-1 and FE-2, the FE estimator is unbiased / consistent.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)**
- 9 Arellano-Bond estimator (a CF + IV estimator)

Cochrane-Orcutt estimator

- The assumption that the firm-specific transitory shock is not serially correlated (and fully unknown to the firm at period t) is quite strong.
- This assumption is testable (Arellano-Bond test for serial correlation). If rejected, this assumption can be relaxed.
- Suppose that we maintain assumption FE-1 but we replace assumption FE-2 with the following.
- **Assumption FE-CO.** The firm-specific transitory shock, u_{it} , follows an Autoregressive-1 process, AR(1):

$$u_{it} = \rho u_{i,t-1} + a_{it}$$

where ρ is a parameter, and a_{it} is not correlated over time and it is realized after the firm chooses the amount of inputs at period t .

Cochrane-Orcutt estimator [2]

- In this model where u_{it} is serially correlated, the standard FE estimator is inconsistent (biased) because $u_{it} - \bar{u}_i$ is correlated with the regressors.
- However, we can define a new version of the FE estimator (Cochrane-Orcutt FE) that is consistent under these conditions and the additional condition that the number of periods T is large.

Cochrane-Orcutt estimator [3]

- Consider the PF at periods t and $t - 1$ under assumption FE-1:

$$y_{it} = \alpha_L \ell_{it} + \alpha_K k_{it} + \eta_i + \delta_t + u_{it}$$

$$y_{it-1} = \alpha_L \ell_{it-1} + \alpha_K k_{it-1} + \eta_i + \delta_{t-1} + u_{it-1}$$

- Multiplying the equation at $t - 1$ by ρ and subtracting it to the equation at period t , we get:

$$\begin{aligned} y_{it} - \rho y_{it-1} &= \alpha_L [\ell_{it} - \rho \ell_{it-1}] + \alpha_K [k_{it} - \rho k_{it-1}] \\ &\quad + [1 - \rho] \eta_i + [\delta_t - \rho \delta_{t-1}] + a_{it} \end{aligned}$$

because $u_{it} - \rho u_{it-1} = a_{it}$.

- This is called a **quasi-difference transformation**.

Cochrane-Orcutt estimator [4]

- The quasi-difference transformation can be written as:

$$y_{it} = \beta_1 y_{i,t-1} + \beta_2 \ell_{it} + \beta_3 \ell_{i,t-1} + \beta_4 k_{it} + \beta_5 k_{i,t-1} + \eta_i^* + \delta_t^* + a_{it} \quad (3)$$

with $\beta_1 = \rho$, $\beta_2 = \alpha_L$, $\beta_3 = -\rho\alpha_L$, $\beta_4 = \alpha_K$, $\beta_5 = -\rho\alpha_K$, and $\eta_i^* = (1 - \rho)\eta_i$, and $\delta_t^* = \delta_t - \rho\delta_{t-1}$.

- Note that given the β parameters we can obtain the parameters ρ , α_L , and α_K . In fact, there are additional (over-identifying restrictions):

$$\rho = \beta_1 = -\beta_3/\beta_2 = -\beta_5/\beta_4$$

$$\alpha_L = \beta_2 = -\beta_3/\beta_1$$

$$\alpha_K = \beta_4 = -\beta_5/\beta_1$$

- Now, under assumption FE-CO, in equation (2), the transitory shock a_{it} is not correlated with the inputs.

Cochrane-Orcutt estimator: Implementation

```
xtreg logy logn logk l.logy l.logn l.logk i.year, fe
```

Fixed-effects (within) regression
Group variable: id

Number of obs = 3,563
Number of groups = 509

R-sq:

within = 0.7825
between = 0.9879
overall = 0.9847

Obs per group:

min = 7
avg = 7.0
max = 7

corr(u_i, Xb) = 0.7191

F(11,3043) = 995.10
Prob > F = 0.0000

logy	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
logn	.4880013	.0166747	29.27	0.000	.4553065	.5206961
logk	.1765454	.0178288	9.90	0.000	.1415877	.2115032
logy L1.	.4039344	.015273	26.45	0.000	.3739879	.4338808
logn L1.	-.0231194	.0192464	-1.20	0.230	-.0608566	.0146179
logk L1.	-.1305487	.0164086	-7.96	0.000	-.1627218	-.0983757

Cochrane-Orcutt estimator: Large T condition

- Consider equation (3) in firm-specific means:

$$\bar{y}_i = \beta_1 \bar{y}_{i(-1)} + \beta_2 \bar{\ell}_i + \beta_3 \bar{\ell}_{i(-1)} + \beta_4 \bar{k}_i + \beta_5 \bar{k}_{i(-1)} + \eta_i^* + \bar{\delta}^* + a_i$$

- And in deviations with respect to firm-specific means:

$$\begin{aligned} (y_{it} - \bar{y}_i) &= \beta_1 [y_{it-1} - \bar{y}_{i(-1)}] + \beta_2 [\ell_{it} - \bar{\ell}_i] + \beta_3 [\ell_{it-1} - \bar{\ell}_{i(-1)}] \\ &\quad + \beta_4 [k_{it} - \bar{k}_i] + \beta_5 [k_{it-1} - \bar{k}_{i(-1)}] \\ &\quad + (\delta_t^* - \bar{\delta}^*) + (a_{it} - \bar{a}_i) \end{aligned}$$

- The FE-Cochrane-Orcutt estimator is applying OLS to this equation.

Cochrane-Orcutt estimator: Large T condition [2]

- **IMPORTANT NOTE:** The FE-Cochrane-Orcutt is consistent (asymptotically unbiased) only when T is large, e.g., larger than 30 or 40 periods.
- Note that under condition FE-OC, we have that $(a_{it} - a_i)$ is not correlated with regressors $[\ell_{it} - \bar{\ell}_i]$, $[\ell_{i,t-1} - \bar{\ell}_{i(-1)}]$, $[k_{it} - \bar{k}_i]$, and $[k_{i,t-1} - \bar{k}_{i(-1)}]$.
- However, even under this condition, we have that $(a_{it} - \bar{a}_i)$ is correlated with regressor $(y_{it-1} - \bar{y}_{i(-1)})$.
- Note that y_{it-1} depends on $a_{i,t-1}$ and that $a_{i,t-1}$ is part of $\bar{a}_i$.
- This correlation goes to zero as T becomes large.

Table of Contents

- 1 Model
- 2 Data
- 3 What Determines Total Factor Productivity?
- 4 Estimation: The Simultaneity Problem
- 5 Simultaneity Problem: Solutions
- 6 Input prices as IVs
- 7 Fixed Effects estimator (a CF estimator)
- 8 Fixed Effects + Cochrane-Orcutt estimator (a CF estimator)
- 9 Arellano-Bond estimator (a CF + IV estimator)

Arellano-Bond estimator

- Assumption FE-2 (or for that matter FE-CO) has two parts:
 - FE-2(a): u_{it} is not serially correlated.
 - FE-2(b): u_{it} is not known to the firm when it decides the amounts of inputs.
- In most applications, the stronger of the two assumptions is FE-2(b).
- We now present a panel data estimator that relaxes assumption FE-2(b).

Arellano-Bond estimator [2]

- We maintain assumptions FE-1, $\omega_{it} = \eta_i + \delta_t + u_{it}$, and FE-2(a), u_{it} is not serially correlated.
- Define the variables in first differences: $\Delta y_{it} = y_{it} - y_{it-1}$; $\Delta \ell_{it} = \ell_{it} - \ell_{it-1}$; etc.
- And consider the PF in first differences (equation at period t minus equation at period $t - 1$):

$$\Delta y_{it} = \alpha_L \Delta \ell_{it} + \alpha_K \Delta k_{it} + \Delta \delta_t + \Delta u_{it}$$

- We have removed the term η_i from the error term, and we can control for the term $\Delta \delta_t$ by including time-dummies.
- But we still have the term Δu_{it} that is correlated with the regressors $\Delta \ell_{it}$ and Δk_{it} .

Arellano-Bond estimator [3]

$$\Delta y_{it} = \alpha_L \Delta \ell_{it} + \alpha_K \Delta k_{it} + \Delta \delta_t + \Delta u_{it}$$

- Consider the following general models for demand of capital and labor inputs:

$$(LD) \ell_{it} = \beta_1^{LD} \ell_{i,t-1} + \beta_2^{LD} k_{i,t-1} + \beta_3^{LD} \omega_{it} + \beta_4^{LD} r_{it}$$

$$(KD) k_{it} = \beta_1^{KD} \ell_{i,t-1} + \beta_2^{KD} k_{i,t-1} + \beta_3^{KD} \omega_{it} + \beta_4^{KD} r_{it}$$

- This means that ℓ_{it} and k_{it} depends on the current and the past histories of the transitory shocks: u_{it} , u_{it-1} , $u_{i,t-2}$, etc.;
- But not on future shocks: u_{it+1} , u_{it+2} , etc.

Arellano-Bond estimator [4]

$$\Delta y_{it} = \alpha_L \Delta \ell_{it} + \alpha_K \Delta k_{it} + \Delta \delta_t + \Delta u_{it}$$

- This implies that $\ell_{i,t-2}$ and $k_{i,t-2}$ are valid instruments in this equation.
- They are **Relevant**: $\Delta \ell_{it}$ and Δk_{it} are correlated with $\ell_{i,t-2}$ and $k_{i,t-2}$.
- They are **not correlated with the error term**:

$$E[\ell_{it-2} \Delta u_{it}] = E[\ell_{it-2} u_{it}] - E[\ell_{it-2} u_{it-1}] = 0$$

Arellano-Bond estimator [5]

- This idea implies many moment restrictions that can be used to estimate α_L , α_K , and $\Delta\delta_t$:

$$E(\ell_{it-j}\Delta u_{it}) = 0, \quad \text{for } t = 3, \dots, T; \text{ and } j \leq t - 2$$

$$E(k_{it-j}\Delta u_{it}) = 0, \quad \text{for } t = 3, \dots, T; \text{ and } j \leq t - 2$$

$$E(y_{it-j}\Delta u_{it}) = 0, \quad \text{for } t = 3, \dots, T; \text{ and } j \leq t - 2$$

- The Arellano-Bond estimator exploits all these restrictions optimally: optimal weighting; optimal Generalized Method of Moments (GMM) estimator.

Arellano-Bond estimator: Implementation

- The command `xtabond2` in Stata implements the Arellano-Bond estimator.
- It can be applied also to the model where u_{it} follows an autoregressive process.